



University Of Barishal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CSE)

LAB FINAL EXAMINATION - 2026

3rd YEAR, 2nd SEMESTER (2021-22)

COURSE CODE: CSE-3212

COURSE TITLE: Mobile App Development Lab

1) Instructions

- Create a **clean, responsive, and accessible** Flutter app with at least **four** screens.
 - Use a **state management** solution (Provider/Riverpod/Bloc/Cubit/GetX).
 - Integrate with **OpenTDB** for categories and questions.
 - You don't need to match the Figma 1:1, but aim to be **design-close** and **consistent**.
Figma: <https://tinyurl.com/cse8-lab-exam-figma>
-

2) Core Features & Screens

2.1 Screen 1 — Welcome

Purpose: First impression and entry point.

UI:

- App title: “**Quizzical**”
- Primary CTA: “**Start Quiz**” (ElevatedButton)

Navigation:

- Welcome → **CategorySelection**

Acceptance Criteria:

- Button press navigates to Category screen.

- Layout adapts to small/large devices.
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2.2 Screen 2 — Category Selection

Purpose: Pick a trivia category visually.

Data: Fetch from `GET https://opentdb.com/api_category.php`

UI:

- Grid/List of **cards** with category names. **You may use icon or use bg color only**
- **Skeleton/shimmer** while loading(optional).
- **Retry banner** on error.

Interaction:

- Tap a card → pass `categoryId` to next screen.

Acceptance Criteria:

- Categories load once and are cached during session.
 - Tapping a category navigates to Quiz Configuration with the selected category id.
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2.3 Screen 3 — Quiz Configuration & Quiz

Purpose: Configure, then play the quiz.

Configuration UI:

- **Amount:** slider/dropdown (1–50; default 10)
- **Difficulty:** dropdown (**Any / Easy / Medium / Hard**)
- **Type:** dropdown (**multiple / boolean**)
- **Start** button → triggers question fetch.

Question Fetch:

`GET https://opentdb.com/api.php`

?amount=<n>&category=<id>&difficulty=<level>&type=<multiple|boolean>

Quiz UI/Logic:

- Show **one question** at a time.
- For **multiple**: 4 shuffled answer buttons.
- For **boolean**: 2 buttons (**True / False**).
- **Timer** per question (e.g., 20–30s). Timeout = incorrect/unanswered, auto-advance.
- Visual feedback (**correct/incorrect**) and **Next** button.
- **Score** and **question counter** update live with a progress bar below appbar.

Acceptance Criteria:

- Start fetch respects chosen **amount**, **category**, **difficulty**, **type**.
 - Answers are **shuffled**; exactly one correct.
 - **Timer** visible and functional; no overlap on navigation.
 - **Loading skeletons** present; errors show retry preserving config.
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2.4 Screen 4 — Results

Purpose: Celebrate result and enable replay.

UI:

- Prominent score: e.g., “**You scored 7/10!**”
- Quick stats: accuracy %, total time.
- CTA: “**Play Again**” → back to Welcome or Category Selection.

Acceptance Criteria:

- Displays correct totals from the just-finished session.

- Replay resets quiz state, preserves last **config** via SharedPreferences.
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3) API Integration

3.1 Categories Endpoint

- URL: https://opentdb.com/api_category.php
- Response (sample):

```
{
  "trivia_categories": [
    { "id": 9, "name": "General Knowledge" },
    { "id": 10, "name": "Entertainment: Books" }
  ]
}
```

3.2 Questions Endpoint

- URL pattern:

```
https://opentdb.com/api.php?amount=<n>&category=<id>&difficulty=<easy|medium|hard>&type=<multiple|boolean>
```

- Response (sample):

```
{
  "response_code": 0,
  "results": [
    {
      "type": "boolean",
      "difficulty": "easy",
      "category": "General Knowledge",
      "question": "When you cry in space, your tears stick to your face.",
      "correct_answer": "True",
      "incorrect_answers": [
        "False"
      ]
    }
  ]
}
```

```
    ]
  },
  {
    "type": "multiple",
    "difficulty": "easy",
    "category": "General Knowledge",
    "question": "How many colors are there in a rainbow?",
    "correct_answer": "7",
    "incorrect_answers": [
      "8",
      "9",
      "10"
    ]
  },
}
```

Note: for category api, it doesn't provide you any icon/image, so you can use different bg colors (variations for each)